

SYDNEY TECHNICAL HIGH SCHOOL

(Established 1911)



TRIAL HIGHER SCHOOL CERTIFICATE

2014

Mathematics Extension 2

General Instructions

- Reading time - 5 minutes
- Working time - 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations
- Diagrams are not drawn to scale
- Start each question on a new page

Total marks - 100

Section 1 - 10 marks

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section 2 - 90 marks

- Attempt Questions 11-16
- Allow about 2 hours and 45 minutes for this section

Name : _____

Teacher : _____

Section 1

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple choice answer sheet for Questions 1-10.

- 1 The graph of $y = g(x)$ is a reflection of the graph of $y = f(x)$ about the x axis.
Which of the following must be true ?

(A) $g(x) = |f(x)|$

(B) $g(x) = -f(x)$

(C) $g(x) = f(-x)$

(D) $g(x) = f^{-1}(x)$

- 2 The slope of the curve $2x^3 - y^2 = 7$ at the point where $y = -3$ is

(A) 4

(B) -1

(C) -2

(D) -4

3 If $z = -2i$, then $|z^2|$ and $\text{Arg}(z^2)$ are respectively

(A) 4 and 0

(B) -4 and π

(C) 4 and π

(D) -4 and $-\pi$

4 If $1 + 2i$ is a root of the polynomial equation $2x^3 + bx^2 + 2x + 20 = 0$,

where b is real, then the value of b is

(A) -3

(B) -2

(C) 0

(D) 2

5 The region bounded by the lines $x = -1$, $y = 1$, $y = -1$ and the curve $x = y^2$

is rotated about the line $x = -2$ to form a solid of revolution.

Which is the correct expression for the volume of this solid ?

(A) $\pi \int_{-1}^1 y^4 - 4y^2 + 3 \, dy$

(B) $\pi \int_{-1}^1 y^4 + 4y^2 + 3 \, dy$

(C) $\pi \int_{-1}^1 y^4 - 4y^2 + 4 \, dy$

(D) $\pi \int_{-1}^1 y^4 + 4y^2 + 4 \, dy$

- 6 The x axis is tangent to an ellipse at the point $(1, 0)$ and the y axis is tangent to the same ellipse at the point $(0, -2)$.

Which one of the following could be the equation of this ellipse ?

(A) $\frac{(x-1)^2}{4} + (y+2)^2 = 1$

(B) $\frac{(x+1)^2}{4} + (y-2)^2 = 1$

(C) $(x+1)^2 + \frac{(y-2)^2}{4} = 1$

(D) $(x-1)^2 + \frac{(y+2)^2}{4} = 1$

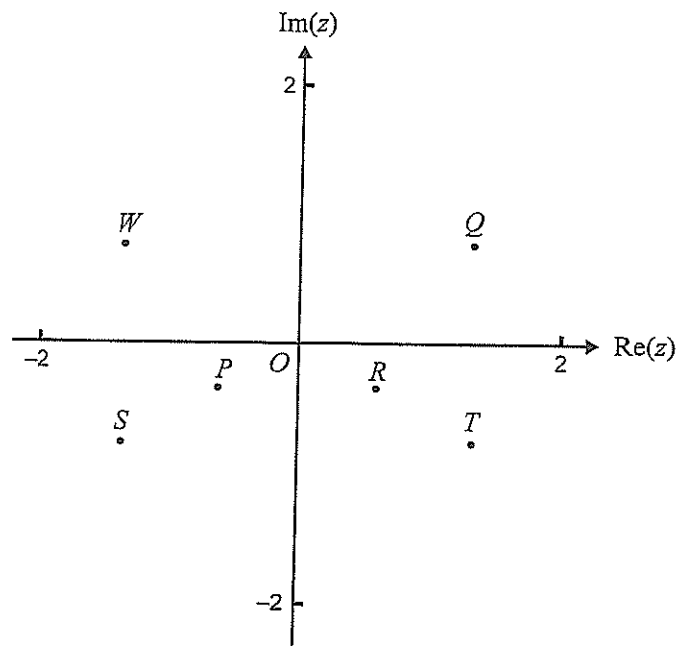
- 7 Which one of the following relations does **not** have a graph that is a straight line passing through the origin?

(A) $z + \bar{z} = 0$

(B) $z = i\bar{z}$

(C) $3 \operatorname{Re}(z) = \operatorname{Im}(z)$

(D) $\operatorname{Re}(z) + \operatorname{Im}(z) = 1$



The point W on the Argand diagram above represents a complex number w where $|w| = 1.5$. The complex number w^{-1} is best represented by the point

- (A) P
- (B) R
- (C) S
- (D) T

9 With a suitable substitution, $\int_0^{\frac{\pi}{6}} \cos^3(2x) dx$ can be expressed as

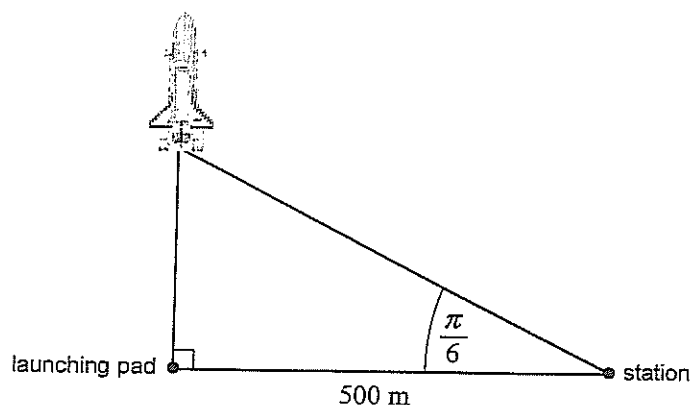
(A) $\frac{1}{2} \int_0^{\frac{\sqrt{3}}{2}} 1 - u^2 du$

(B) $\frac{1}{2} \int_0^{\frac{1}{2}} 1 - u^2 du$

(C) $2 \int_0^{\frac{\sqrt{3}}{2}} u^2 - 1 du$

(D) $2 \int_0^{\frac{1}{2}} 1 - u^2 du$

10



An ascending space shuttle rises vertically from a launching pad. As it rises the shuttle is tracked from a station at ground level 500 metres away. When the angle of elevation of the shuttle is $\frac{\pi}{6}$ radians from the horizontal, and is increasing at a rate of 0.5 radians per second, the speed of the shuttle is closest to

- (A) 144 metres per second
- (B) 289 metres per second
- (C) 333 metres per second
- (D) 577 metres per second

Section 2

90 marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section

Start each question on a new page.

In Questions 11-16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)

a) Let $z = -1 + i\sqrt{3}$ and $w = \sqrt{3} + i$

i) Find $z \div \bar{w}$, in simplest $a + ib$ form.

2

ii) Find $\text{Arg}(z)$.

1

iii) Find $|z|$.

1

iv) Find the smallest value of n , given that $n > 1$,

1

such that $\text{Arg}(z) = \text{Arg}(z^n)$

b) Find $\int \frac{1}{2x^2 - x} dx$

3

c) Find $\int x \sec^2 x dx$

3

d) Consider the polynomial $P(z) = z^3 + 9z^2 + 28z + 20$.

2

Given that $P(-1) = 0$, fully factorise $P(z)$ over the complex field.

e) Without the use of calculus, sketch the curve $9y^2 = x(3 - x)^2$.

2

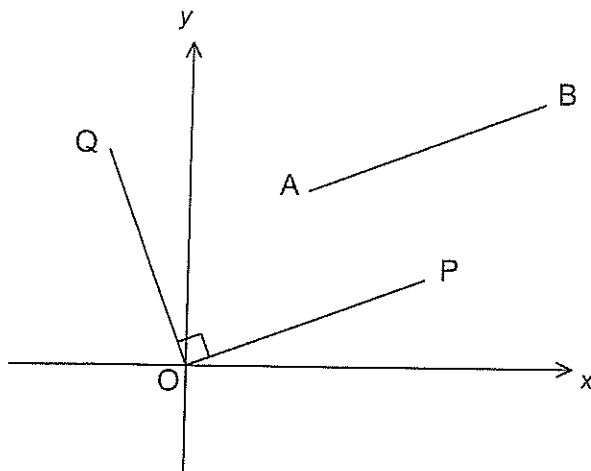
Question 12 (15 marks) [Start a new page](#)

- a) Find the equation of the hyperbola with foci at the points $(\pm 8, 0)$ and vertices at the points $(\pm 5, 0)$ 2
- b) Solve the equation $x^4 - 5x^3 - 9x^2 + 81x - 108 = 0$ 3
given that it has a triple root.
- c) If α, β and γ are the roots of the equation $x^3 - 5x + 3 = 0$ 2
find the polynomial equation with roots $\frac{1}{\alpha+1}, \frac{1}{\beta+1}$ and $\frac{1}{\gamma+1}$.
- d) i) Given that $I_n = \int_0^{\frac{\pi}{2}} x^n \sin x \, dx$ 4
show that $I_n = n \left(\frac{\pi}{2}\right)^{n-1} - n(n-1) I_{n-2}$
- ii) Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{2}} x^4 \sin x \, dx$. 2

Question 12 (continued)

e)

2



In the Argand diagram above, intervals AB , OP and OQ are equal in length.

OP is parallel to AB and angle $POQ = 90^\circ$.

If A and B represent the complex numbers $3 + 5i$ and $9 + 8i$ respectively,
find the complex number which is represented by Q .

Question 13 (15 marks) Start a new page

- a) Solve for x , $\tan^{-1}3x - \tan^{-1}2x = \tan^{-1}\frac{1}{5}$ 2
- b) Use the substitution $x = 10 \sin \theta$ to evaluate $\int_0^5 \sqrt{100 - x^2} \, dx$ 4
- c) Find the volume of a solid whose base is a triangle with vertices $(0, 0)$, $(2, 0)$ and $(0, 2)$ and whose cross sections perpendicular to the base and parallel to the y axis are semi-circles. 4
- d) i) Show that the equation of the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ 2
at $P(a \sec \theta, b \tan \theta)$ is given by $\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$.
- ii) If the tangent to $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point $P(a \sec \theta, b \tan \theta)$ 3
cuts the x axis at A and the y axis at B , show that $\frac{PA}{PB} = \sin^2 \theta$

Question 14 (15 marks) Start a new page

a) Consider the function $f(x) = \frac{x^3+4}{x^2}$.

i) Find the coordinates of any stationary points on $y = f(x)$ and determine their nature. 2

ii) For what values of x is the curve $y = f(x)$ is concave up? 2

iii) Sketch the curve $y = f(x)$ showing any important features. 2

iv) On a separate diagram, and without the use of further calculus, sketch the curve $y = \frac{1}{f(x)}$ 2

b) A particle P is projected vertically upwards from the surface of the Earth with initial velocity u . The acceleration due to gravity at any point on its path is given by $-\frac{k}{x^2}$, where x is the distance of the particle from the centre of the Earth and k is a constant.

i) Explain why $k = gR^2$, where R is the radius of the Earth and g is the acceleration due to gravity at the Earth's surface. 1

ii) Neglecting air resistance, show that the velocity v of the particle is given by 3

$$v^2 = u^2 - 2gR^2\left(\frac{1}{R} - \frac{1}{x}\right)$$

iii) If the initial velocity of the particle is given by $u = \sqrt{2gR}$, show that the time taken to reach a height $3R$ above the Earth's surface is given by $\frac{7\sqrt{2R}}{3\sqrt{g}}$. 3

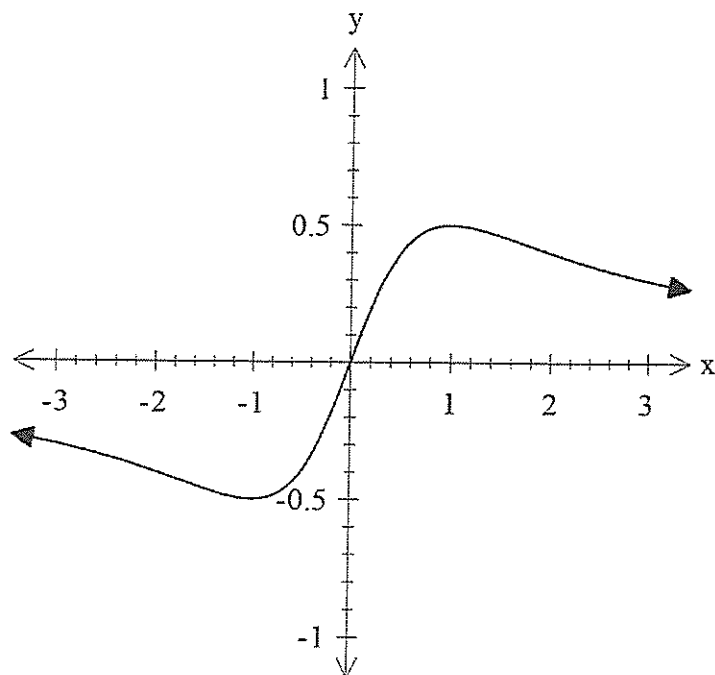
Question 15 (15 marks) Start a new page

a) For the relation $x^3 + y^3 = 6xy$, find $\frac{dy}{dx}$ in terms of x and y . 2

b) Sketch the region on the Argand diagram defined by 2

$$(z - 3 + i)(\bar{z} - 3 - i) \leq 9$$

c) The following is a graph of $y = \frac{x}{x^2+1}$

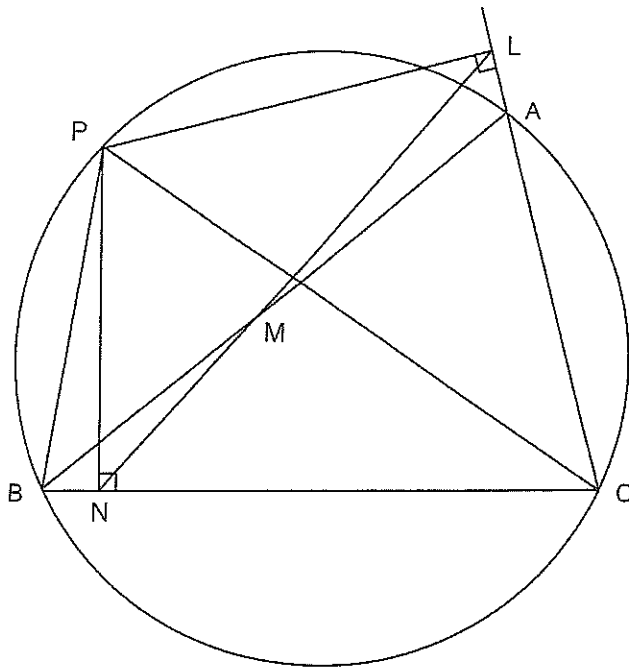


The area bounded by the curve $y = \frac{x}{x^2+1}$, the line $x = 1$ and the x axis 4

is rotated about the y axis. Using the method of cylindrical shells, or otherwise, find the volume of the solid of revolution formed.

Question 15 (continued)

d)



ABC is an acute angled triangle inscribed in a circle. P is a point on the minor arc AB such that PL is perpendicular to CA (produced) and PN is perpendicular to BC .

LN cuts AB at M .

Copy or trace the diagram into your Writing Booklet.

- | | |
|--|---|
| i) Explain why $PNCL$ is a cyclic quadrilateral. | 1 |
| ii) Show that $\angle PBM = \angle PNM$. | 3 |
| iii) Show that PM is perpendicular to AB . | 3 |

Question 16 (15 marks) Start a new page

a) If w is a complex root of the equation $x^3 = 1$.

i) Show that the other complex root is w^2 . 1

ii) Show that $1 + w + w^2 = 0$. 1

iii) Find in its simplest form, the cubic equation whose roots are 3

$3, 2w + w^2$ and $2w^2 + w$.

b) i) Given that for $k > 0$, $2k + 3 > 2\sqrt{(k+1)(k+2)}$, 4

Use mathematical induction to show that $\sum_{r=1}^n \frac{1}{\sqrt{r}} > 2(\sqrt{n+1} - 1)$
for all positive integers n .

ii) Use the graph of $y = \frac{1}{\sqrt{x}}$ to show that 3

$$\sum_{r=1}^n \frac{1}{\sqrt{r}} < 1 + \int_1^n \frac{1}{\sqrt{x}} dx$$

iii) Hence show that $198 < \sum_{r=1}^{10000} \frac{1}{\sqrt{r}} < 199$ 3

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

EXT 2 SOLUTIONS 2014 TRIAL HSC

1. B

2. D

3. C

4. C

5. B

6. D

7. D

8. A

9. A

10. C

$$\begin{aligned}
 11. \quad a) \quad i) \quad & \frac{-1 + i\sqrt{3}}{\sqrt{3} - i} \times \frac{\sqrt{3} + i}{\sqrt{3} + i} \\
 &= \frac{-\sqrt{3} - i + 3i - \sqrt{3}}{4} \\
 &= \frac{-\sqrt{3} + i}{2}
 \end{aligned}$$

$$ii) \quad \text{Arg}(z) = \frac{2\pi}{3}$$

$$iii) \quad |z| = 2$$

$$iv) \quad 2m\pi + \frac{2\pi}{3} = n \times \frac{2\pi}{3}$$

$$\therefore n = 4$$

$$b) \quad \int \frac{1}{2x^2 - x} dx$$

$$\frac{1}{x(2x-1)} = \frac{A}{x} + \frac{B}{2x-1}$$

$$= \int \frac{2}{2x-1} - \frac{1}{x} dx$$

$$\therefore 1 = A(2x-1) + Bx$$

sub $x=0$

$$1 = -A$$

$$= \ln(2x-1) - \ln x + C$$

$$\Rightarrow A = -1$$

$$= \ln\left(\frac{2x-1}{x}\right) + C$$

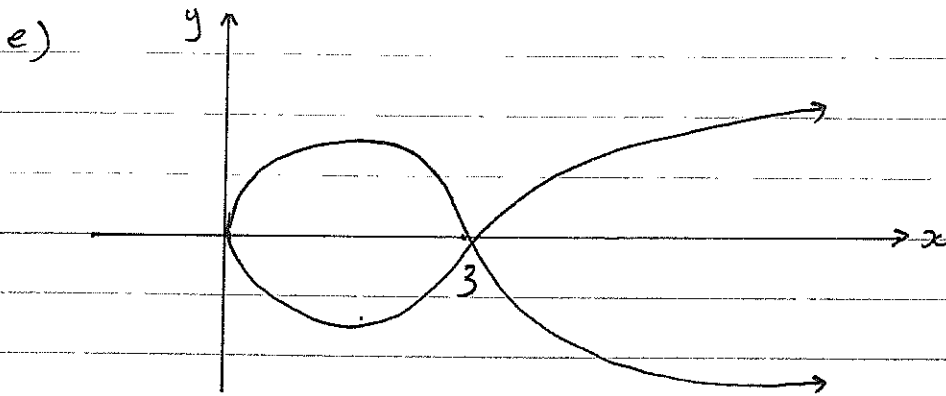
$$\therefore B = 2$$

$$c) \int x \sec^2 x \, dx \quad \begin{array}{ll} u = x & v = \tan x \\ u' = 1 & v' = \sec^2 x \end{array}$$

$$= x \tan x - \int \tan x \, dx$$

$$= x \tan x + \ln(\cos x) + C$$

$$\begin{aligned} d) \quad P(z) &= z^3 + 9z^2 + 28z + 20 \\ &= (z+1)(z^2 + 8z + 20) \\ &= (z+1)((z+4)^2 + 4) \\ &= (z+1)(z+4+2i)(z+4-2i) \end{aligned}$$



$$\boxed{12} \quad a) \quad \begin{array}{ll} S(\pm 8, 0) & ae = 8 \\ A(\pm 5, 0) & a = 5 \\ & e = \frac{8}{5} \end{array}$$

$$\begin{aligned} b^2 &= a^2(e^2 - 1) \\ &= 25\left(\frac{64}{25} - 1\right) \\ &= 39 \end{aligned}$$

$$\therefore \frac{x^2}{25} - \frac{y^2}{39} = 1$$

$$b) \quad x^4 - 5x^3 - 9x^2 + 81x - 108 = 0$$

$$P'(x) = 4x^3 - 15x^2 - 18x + 81$$

$$P''(x) = 12x^2 - 30x - 18$$

$$\therefore 6(2x^2 - 5x - 3) = 0$$

$$6(2x+1)(x-3) = 0$$

$$x = -\frac{1}{2}, 3$$

$$P'(3) = 0 \quad P'(-\frac{1}{2}) \neq 0$$

$$\therefore \text{roots are } 3, 3, 3, \alpha$$

$$\text{sum of roots } 3 + 3 + 3 + \alpha = 5 \quad \left(-\frac{b}{a}\right)$$

$$\alpha = -4$$

$$\therefore \text{roots } x = 3, 3, 3, -4$$

$$c) \quad \text{let } y = \frac{1}{x+1}$$

$$\therefore x = \frac{1}{y} - 1$$

$$\therefore \text{required equation } P\left(\frac{1}{y} - 1\right) = 0$$

$$\left(\frac{1}{y} - 1\right)^3 - 5\left(\frac{1}{y} - 1\right) + 3 = 0$$

$$\frac{1}{y^3} - \frac{3}{y^2} + \frac{3}{y} - 1 - \frac{5}{y} + 5 + 3 = 0$$

$$\frac{1}{y^3} - \frac{3}{y^2} - \frac{2}{y} + 7 = 0 \quad (\times y^3)$$

$$1 - 3y - 2y^2 + 7y^3 = 0$$

$$d) \quad i) \quad I_n = \int_0^{\frac{\pi}{2}} x^n \sin x \, dx \quad \begin{array}{l} u = x^n \\ u' = n(x)^{n-1} \end{array} \quad \begin{array}{l} v = -\cos x \\ v' = \sin x \end{array}$$

$$= -x^n \cos x \Big|_0^{\frac{\pi}{2}} + n \int_0^{\frac{\pi}{2}} x^{n-1} \cos x \, dx$$

$$\begin{array}{l} u = x^{n-1} \\ u' = (n-1)x^{n-2} \end{array} \quad \begin{array}{l} v = \sin x \\ v' = \cos x \end{array}$$

$$= 0 + n \left[x^{n-1} \sin x \Big|_0^{\frac{\pi}{2}} - (n-1) \int_0^{\frac{\pi}{2}} x^{n-2} \sin x \, dx \right]$$

$$= n \left(\frac{\pi}{2} \right)^{n-1} - n(n-1) I_{n-2}$$

$$ii) \quad I_4 = \int_0^{\frac{\pi}{2}} x^4 \sin x \, dx$$

$$= 4 \left(\frac{\pi}{2} \right)^3 - 4 \times 3 I_2$$

$$= 4 \left(\frac{\pi}{2} \right)^3 - 12 \left[2 \left(\frac{\pi}{2} \right)^2 - 2 \times 1 I_0 \right]$$

$$= 4 \left(\frac{\pi}{2} \right)^3 - 24 \left(\frac{\pi}{2} \right)^2 + 24 \int_0^{\frac{\pi}{2}} \sin x \, dx$$

$$= \pi^3 - 12\pi^2 + 24 (-\cos x) \Big|_0^{\frac{\pi}{2}}$$

$$= \pi^3 - 12\pi^2 + 24 (0 - -1)$$

$$= \pi^3 - 12\pi^2 + 24$$

$$e) \quad P \text{ represents } 6 + 3i$$

$$\therefore Q \text{ represents } i(6 + 3i)$$

$$= -3 + 6i$$

13) a) $\tan^{-1} 3x - \tan^{-1} 2x = \tan^{-1} \frac{1}{5}$

$$\tan(\tan^{-1} 3x - \tan^{-1} 2x) = \frac{1}{5}$$

$$\frac{3x - 2x}{1 + 6x^2} = \frac{1}{5}$$

$$6x^2 - 5x + 1 = 0$$

$$(3x - 1)(2x - 1) = 0$$

$$x = \frac{1}{3}, \frac{1}{2}$$

b) $\int_0^5 \sqrt{100 - x^2} dx$

$$x = 10 \sin \theta$$

$$dx = 10 \cos \theta d\theta$$

$$x = 0 \quad \theta = 0$$

$$x = 5 \quad \theta = \frac{\pi}{6}$$

$$= \int_0^{\frac{\pi}{6}} \sqrt{100 - 100 \sin^2 \theta} \cdot 10 \cos \theta d\theta$$

$$= 100 \int_0^{\frac{\pi}{6}} \sqrt{\cos^2 \theta} \cos \theta d\theta$$

$$= 100 \int_0^{\frac{\pi}{6}} \cos^2 \theta d\theta$$

$$= \frac{100}{2} \int_0^{\frac{\pi}{6}} 1 + \cos 2\theta d\theta$$

$$= 50 \left(\theta + \frac{1}{2} \sin 2\theta \right) \Big|_0^{\frac{\pi}{6}}$$

$$= 50 \left[\left(\frac{\pi}{6} + \frac{1}{2} \sin \frac{\pi}{3} \right) - (0) \right]$$

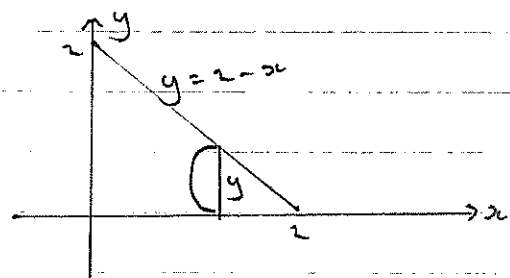
$$= \frac{25\pi}{3} + \frac{25\sqrt{3}}{2}$$

c) $V = \frac{\pi}{8} \int_0^2 y^2 dx$

$$= \frac{\pi}{8} \int_0^2 (2-x)^2 dx$$

$$= -\frac{\pi}{8} \cdot \frac{1}{3} [2-x]^3 \Big|_0^2$$

$$= \frac{\pi}{3} \text{ cu. units}$$



$$A(x) = \frac{1}{2} \pi r^2$$

$$= \frac{1}{2} \pi \left(\frac{1}{4} y \right)^2$$

$$d) \quad i) \quad \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} - \frac{2y}{b^2} \cdot y' = 0$$

$$y' = \frac{b^2 x}{a^2 y}$$

$$\therefore m_T = \frac{b^2 a \sec \theta}{a^2 b \tan \theta}$$

$$= \frac{b \sec \theta}{a \tan \theta}$$

$$\therefore y - b \tan \theta = \frac{b \sec \theta}{a \tan \theta} (x - a \sec \theta)$$

$$ay \tan \theta - ab \tan^2 \theta = bx \sec \theta - ab \sec^2 \theta$$

$$bx \sec \theta - ay \tan \theta = ab (\sec^2 \theta - \tan^2 \theta)$$

$$bx \sec \theta - ay \tan \theta = ab$$

$$\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$$

$$ii) \quad M(0, b \tan \theta)$$

$$B(0, -\frac{b}{\tan \theta})$$

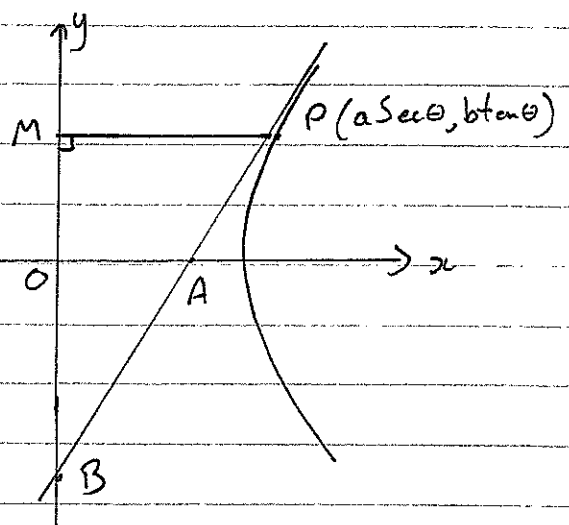
$$\frac{PA}{PB} = \frac{MO}{MB} \quad (\text{ratio of intercepts})$$

$$= \frac{b \tan \theta}{b \tan \theta + \frac{b}{\tan \theta}}$$

$$= \frac{b \tan^2 \theta}{b (\tan^2 \theta + 1)}$$

$$= \frac{\tan^2 \theta}{\sec^2 \theta}$$

$$= \sin^2 \theta$$



14

a) $f(x) = \frac{x^3 + 4}{x^2}, \quad x \neq 0$

1) $f'(x) = \frac{x^2(3x^2) - (x^3 + 4)(2x)}{x^4}$

$= \frac{3x^4 - 2x^4 - 8x}{x^4}$

$= \frac{x^4 - 8x}{x^3}$

$= \frac{x^3 - 8}{x^3}$

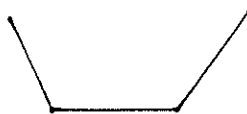
st. pts when $y' = 0$
 $x^3 - 8 = 0$

$x = 2$

$\therefore y = 3$

test

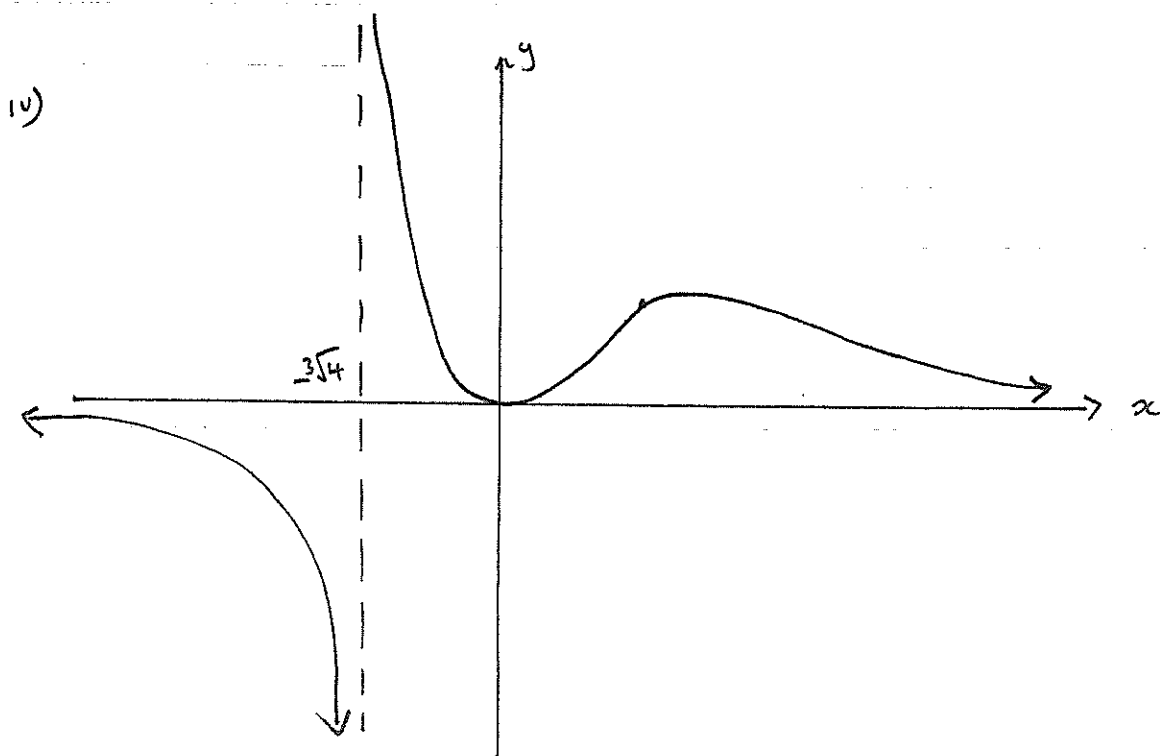
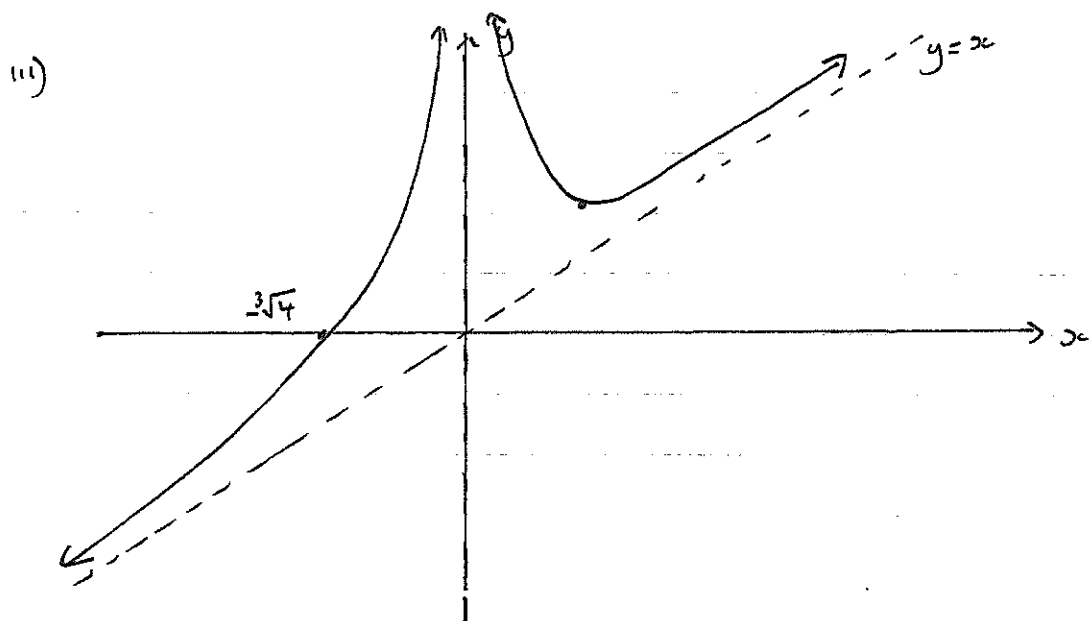
x	2^-	2	2^+
y'	ve	0	ve



$\therefore$ min at $(2, 3)$

$$\begin{aligned}
 \text{ii)} \quad f''(x) &= \frac{x^3(3x^2) - (x^3 - 8)3x^2}{x^6} \\
 &= \frac{3x^5 - 3x^5 + 24x^2}{x^6} \\
 &= \frac{24}{x^4}
 \end{aligned}$$

which is positive for all x , except $x=0$
 $\therefore$ concave up for all $x, x \neq 0$.



$$b) \quad i) \quad \ddot{x} = -\frac{k}{x^2}$$

$$\text{when } x = R \quad \ddot{x} = -g$$

$$\therefore k = R^2 g$$

$$ii) \quad \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -k x^{-2}$$

$$\frac{1}{2} v^2 = k x^{-1} + C$$

$$\text{when } x = R \quad v = u$$

$$\therefore \frac{1}{2} u^2 - \frac{k}{R} = C$$

$$\frac{1}{2} v^2 = \frac{1}{2} u^2 - \frac{k}{R} + \frac{k}{x} \quad \text{but } k = R^2 g$$

$$v^2 = u^2 - \frac{2R^2 g}{R} + \frac{2R^2 g}{x}$$

$$v^2 = u^2 - 2gR^2 \left(\frac{1}{R} - \frac{1}{x} \right)$$

$$iii) \quad \text{If } u = \sqrt{2gR}$$

$$v^2 = 2gR - 2gR + \frac{2gR^2}{x}$$

$$v = \frac{\sqrt{2gR^2}}{\sqrt{x}}$$

$$\frac{dx}{dt} = \frac{\sqrt{2gR^2}}{\sqrt{x}}$$

$$\frac{dt}{dx} = \frac{x^{\frac{1}{2}}}{\sqrt{2gR^2}}$$

$$t = \frac{2x^{\frac{3}{2}}}{3\sqrt{2gR^2}} + C$$

$$\text{when } x = R, \quad t = 0$$

$$\therefore C = -\frac{2R^{\frac{3}{2}}}{3\sqrt{2gR^2}}$$

$$\therefore t = \frac{2x^{\frac{3}{2}}}{3\sqrt{2gR^2}} - \frac{2R^{\frac{3}{2}}}{3\sqrt{2gR^2}} \quad \text{when } x = 4R$$

$$t = \frac{2(4R)^{\frac{3}{2}}}{3\sqrt{2gR^2}} - \frac{2R^{\frac{3}{2}}}{3\sqrt{2gR^2}}$$

$$t = \frac{16\sqrt{R}}{3\sqrt{2g}} - \frac{2\sqrt{R}}{3\sqrt{2g}}$$

$$= \frac{14\sqrt{R}}{3\sqrt{2g}}$$

$$= \frac{7\sqrt{2R}}{3\sqrt{g}}$$

15 a) $3x^2 + 3y^2 \frac{dy}{dx} = 6y + 6x \frac{dy}{dx}$

$$3x^2 - 6y = \frac{dy}{dx} (6x - 3y^2)$$

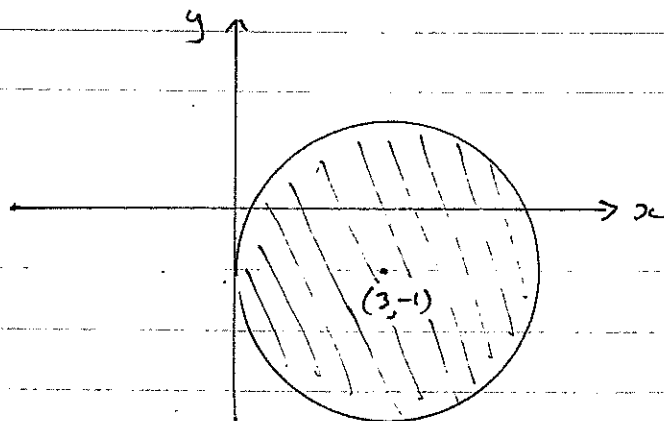
$$\frac{dy}{dx} = \frac{3x^2 - 6y}{6x - 3y^2}$$

$$= \frac{x^2 - 2y}{2x - y^2}$$

b) $(x+iy-3+i)(x-iy-3-i) \leq 9$

$$((x-3) + i(y+1))((x-3) - i(y+1)) \leq 9$$

$$(x-3)^2 + (y+1)^2 \leq 9$$



$$c) \quad V = \int_0^1 2\pi x \left(\frac{x}{x^2+1} \right) dx$$

$$= 2\pi \int_0^1 \frac{x^2}{x^2+1} dx$$

$$= 2\pi \int_0^1 \frac{x^2+1-1}{x^2+1} dx$$

$$= 2\pi \int_0^1 1 - \frac{1}{x^2+1} dx$$

$$= 2\pi \left[x - \tan^{-1} x \right]_0^1$$

$$= 2\pi \left[(1 - \tan^{-1} 1) - (0 - \tan^{-1} 0) \right]$$

$$= 2\pi \left(1 - \frac{\pi}{4} \right) \text{ cu units}$$

d) i) $\angle PNC = \angle PLC = 90^\circ$ (given)

$\therefore$ PNCL is cyclic (opposite angles supplementary)

ii) let $\angle PBM = x$

$\therefore \angle PCA = x$ (angles at circumference equal)

$\therefore \angle PNL = x$ (angles at circumference equal, PNCL cyclic)

$\therefore \angle PBM = \angle PNM$

iii) PMNB is cyclic (PM subtends equal angles at B and N)

$\therefore \angle PMB = \angle PNB$ (PB subtends equal angles at M and N)

but $\angle PNB = 90^\circ$ (given)

$\therefore \angle PMB = 90^\circ$

$\therefore PM \perp AB$

16 a) i) substitute w^2 into x^3 gives

$$(w^2)^3 = (w^3)^2 \\ = 1$$

$\therefore w^2$ is a solution of $x^3 = 1$ if w is a solution.

ii) 3 roots of $x^3 = 1$ are $1, w$ and w^2

$$\therefore \text{sum of roots} = -\frac{b}{a}$$

$$\therefore 1 + w + w^2 = 0$$

iii)

polynomial in form $(x-3)(x^2 - S_1x + S_2) = 0$

$$\begin{aligned} \text{where } S_1 &= 2w + w^2 + 2w^2 + w \\ &= 3w + 3w^2 \\ &= 3(w + w^2) \\ &= -3 \end{aligned}$$

$$\begin{aligned} S_2 &= (2w + w^2)(2w^2 + w) \\ &= 4w^3 + 2w^2 + 2w^4 + w^3 \\ &= 5w^3 + 2w^2 + 2w^4 \quad (w^4 = w^3 \times w) \\ &= 5 + 2w^2 + 2w \\ &= 3 + 2(1 + w^2 + w) \\ &= 3 \end{aligned}$$

$\therefore$ required polynomial is $(x-3)(x^2 + 3x + 3) = 0$

$$x^3 + 3x^2 + 3x - 3x^2 - 9x - 9 = 0$$

$$x^3 - 6x - 9 = 0$$

b) 1) Step 1: test $n=1$ LHS = $\frac{1}{\sqrt{1}}$ RHS = $2(\sqrt{2}-1)$
 $= 1$ $= 2\sqrt{2}-2$
 ≈ 0.828

$\therefore$ true for $n=1$

Step 2: assume true for $n=k$

i.e. $\sum_{r=1}^k \frac{1}{\sqrt{r}} > 2(\sqrt{k+1}-1)$

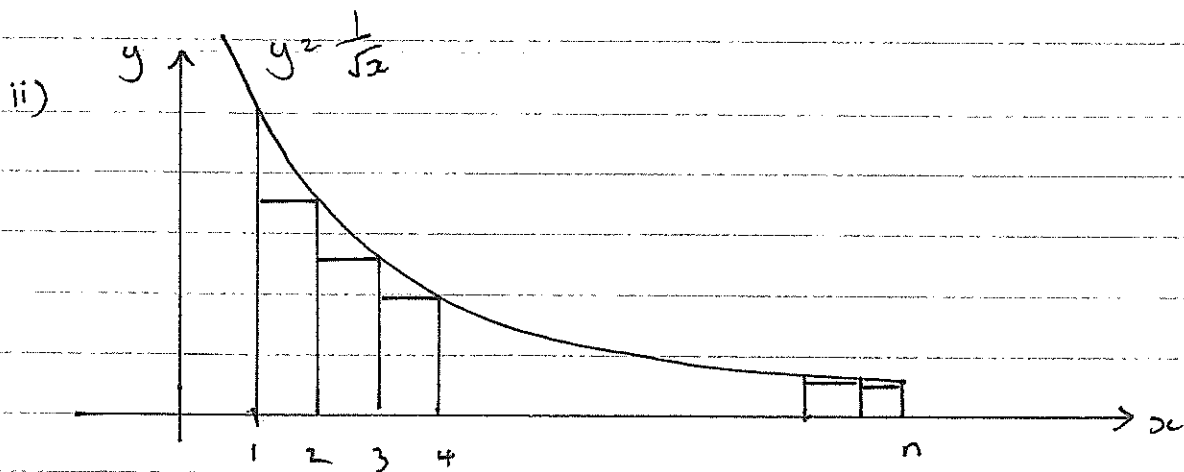
show for $n=k+1$

$$\begin{aligned} \sum_{r=1}^{k+1} \frac{1}{\sqrt{r}} &= \sum_{r=1}^k \frac{1}{\sqrt{r}} + \frac{1}{\sqrt{k+1}} \\ &> 2(\sqrt{k+1}-1) + \frac{1}{\sqrt{k+1}} \\ &= 2\sqrt{k+1} - 2 + \frac{1}{\sqrt{k+1}} \\ &= \frac{2(k+1)}{\sqrt{k+1}} + \frac{1}{\sqrt{k+1}} - 2 \\ &= \frac{2k+3}{\sqrt{k+1}} - 2 \\ &> \frac{2\sqrt{(k+1)(k+2)}}{\sqrt{k+1}} - 2 \\ &= 2(\sqrt{k+2}-1) \end{aligned}$$

which is the required result

$\therefore$ true for $n=k+1$ if true for $n=k$

Step 3: As true for $n=1$, result is true for $n=1+1$, i.e. $n=2$
 As true for $n=2$, result is true for $n=2+1$, i.e. $n=3$
 and so on for all positive integers n .



Area under curve from $x=1$ to n is given by

$$\int_1^n \frac{1}{\sqrt{x}} dx$$

Area of rectangles is given by

$$1 \times \frac{1}{\sqrt{2}} + 1 \times \frac{1}{\sqrt{3}} + 1 \times \frac{1}{\sqrt{4}} + \dots + 1 \times \frac{1}{\sqrt{n}}$$

but area of rectangles $<$ area under curve

$$\therefore \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{n}} < \int_1^n \frac{1}{\sqrt{x}} dx$$

$$\therefore \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{n}} < 1 + \int_1^n \frac{1}{\sqrt{x}} dx$$

$$\therefore \sum_{r=1}^n \frac{1}{\sqrt{r}} < 1 + \int_1^n \frac{1}{\sqrt{x}} dx$$

$$\text{iii) } 2(\sqrt{n+1} - 1) < \sum_{r=1}^n \frac{1}{\sqrt{r}} < 1 + \int_1^n \frac{1}{\sqrt{x}} dx \quad \begin{array}{l} \text{from (i)} \\ \text{and (ii)} \end{array}$$

$$\begin{aligned} & \text{Let } n = 10000 \\ & 2(\sqrt{10001} - 1) < \sum_{r=1}^{10000} \frac{1}{\sqrt{r}} < 1 + \int_1^{10000} \frac{1}{\sqrt{x}} dx \\ & 198.0099 < \sum_{r=1}^{10000} \frac{1}{\sqrt{r}} < 1 + [2\sqrt{x}]_1^{10000} \end{aligned}$$

$$198 < \sum_{r=1}^{10000} \frac{1}{\sqrt{r}} < 1 + 2(\sqrt{10000} - 1)$$

$$198 < \sum_{r=1}^{10000} \frac{1}{\sqrt{r}} < 199$$